

Advancing GeoAI for Disaster Resilience:
From Damage Assessment to Responsible Decision Support

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Committee members:

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"Seeing, describing, and governing disasters — beyond the boundaries of the screen, let the real world become the stage for intelligence." --Yifan Yang

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Outline

- Background
- Current Challenges
- Conceptual Framework
- Objectives
- Case Studies
- Research Schedule

Background

Disaster resilience is a geographical issue.

(1) Rooted in Space:

Driven by vulnerability, exposure, and human-environment dynamics.

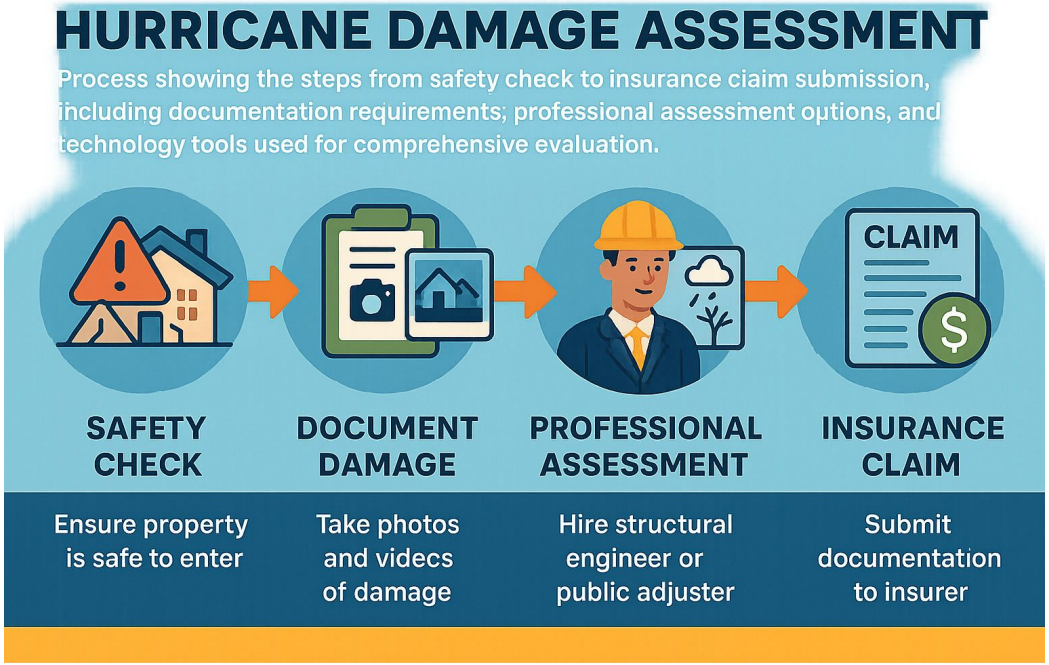
(2) Geographic Inquiry:

Moving beyond *what happened* to *where*, *how severe*, *who was affected*, and *how to allocate resources*.



Background

Disaster assessment is a critical task for building disaster resilience.





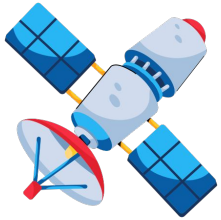
Source: Federal Emergency Management Agency (FEMA). | GAO-25-108598



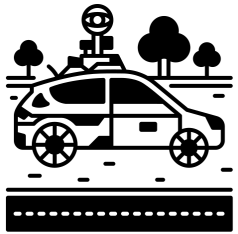
Image Source: https://insuranceclaimrecoverysupport.com/hurricane-damage-assessment/

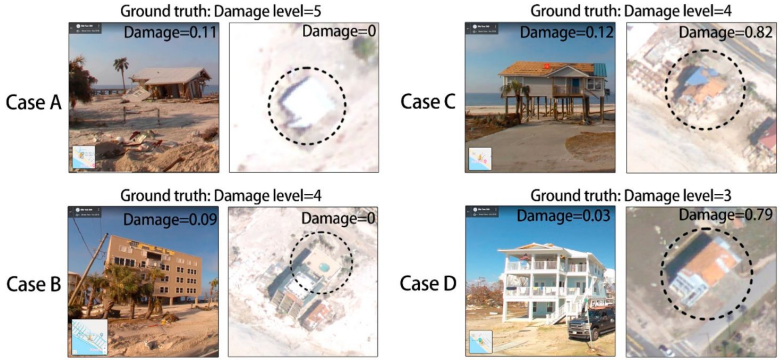
Background

Geospatial data is important for studying disaster resilience.





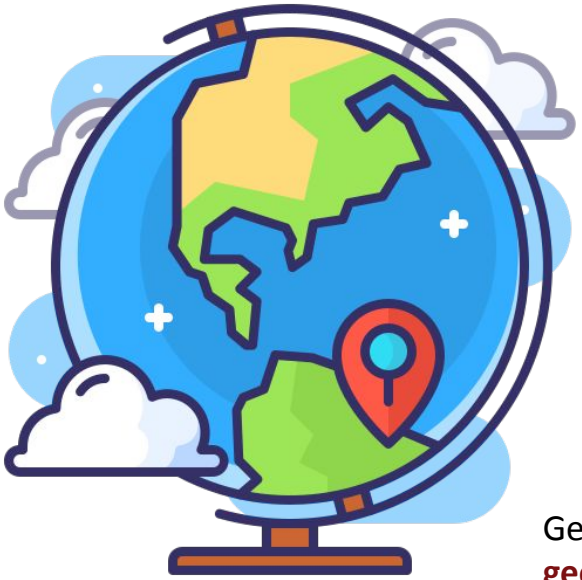




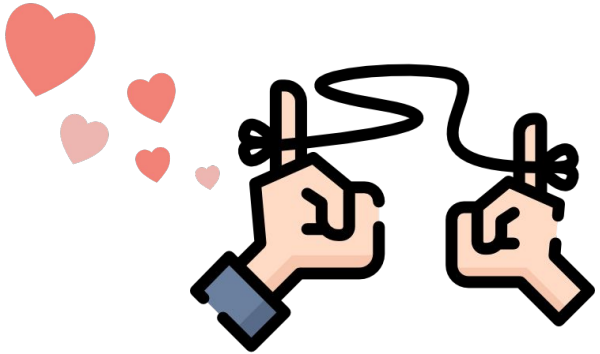


Background

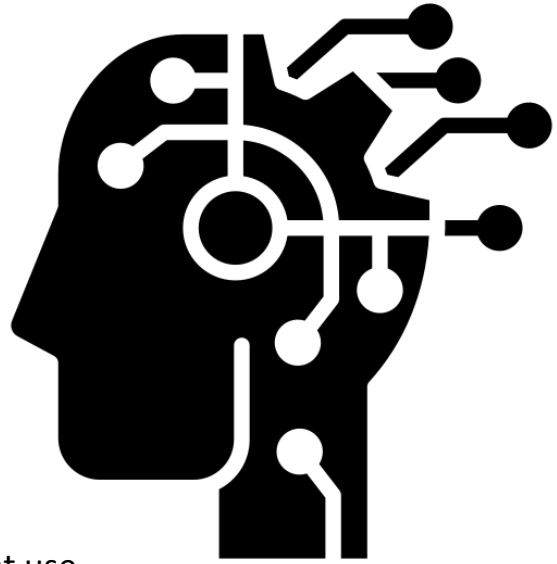
The Engine: GeoAI for Disaster Resilience



Geography



Artificial Intelligence



GeoAI refers to **spatially explicit AI** methods that use **geographic data and spatial context** to **create** intelligent geographic information, knowledge, and services.

Janowicz, K., Gao, S., McKenzie, G., Hu, Y., & Bhaduri, B. (2020). GeoAI: spatially explicit artificial intelligence techniques for geographic knowledge discovery and beyond. International Journal of Geographical Information Science, 34(4), 625-636.

Background

Evolution of GeoAI Paradigms for Disaster Resilience.



Closed-ended questions with labeled answers

(1) Is this building damaged?
(2) What is the extent of the damage—none/minor/severe/total destruction?

Open-ended/Constructive Questions

(1) Why was this location assessed as severely damaged?
(2) What actions should rescue personnel take here?"

Plan, act, and converse

(1) Plan the analysis, call the spatial tools, and show me the evidence behind every claim.

Plan analytical tasks, invoke spatial analysis tools, interact with users — but need accountability.

Background--A one-sentence summary



Leveraging evolving and advancing GeoAI across discriminative models, generative methods, and intelligent agents to achieve a deeper understanding of disaster resilience and enable more scientifically grounded decision-making.

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Current Challenges

1. **Place is only partially observed.**
   This is the problem of seeing: most models use a single post-event view of a single event, discard the pre-disaster baseline, and treat disagreement between views or models as noise rather than information. **See better**

2. **Place-specific disasters require rapid assessment without event-specific training.**
   This is the problem of rapid response: models trained on one hurricane transfer poorly to another, responders cannot wait for new labels and retraining, and overconfident errors can misdirect a rescue team. **Respond faster**

3. **Place is difficult to anticipate, inspect, and communicate.**
   This is the problem of preparedness and delivery: current pipelines end with a label or a map, and autonomous GeoAI agents that could deliver richer intelligence lack accountability, evidence lineage, and refusal when evidence is insufficient. **Prepare and deliver better**



Conceptual Framework

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Challenges

Challenge 1
Place Is Only Partially Observed

Challenge 2
Place-Specific Disasters Require Rapid Assessment

Challenge 3
Place Is Difficult to Anticipate, Inspect, and Communicate

Evolution Stages

Eolving GeoAI for Disaster Resilience



Stage 1
Discriminative Damage Assessment
A single hurricane seen from a single, incomplete ground-level view.





Stage 2
Generative Multi-View Understanding
Many views, regions, and hazard types without event-specific training.





Stage 3
Agentic Decision Support
Anticipating, inspecting, and communicating place-based impacts.



Insights

Insight 1
More complete and more reliable hyperlocal damage observation
[Case 1-2]

Insight 2
Rapid, interpretable, zero-shot assessment across views, regions, and hazards
[Cases 3-5]

Insight 3
Preparedness and accountable place-based decision support
[Cases 6-7]

Objectives

Objective 1 → *addresses Challenge 1* → ***Case 1,2***

Assess damage more completely by observing the same place across overhead, ground-level, and pre-disaster views, and by treating disagreement between views and models as evidence to arbitrate rather than noise to discard. **See better**

Objective 2 → *addresses Challenge 2* → ***Case 3, 4, 5***

Develop damage assessment methods that transfer across events without new labels, retraining, or event-specific models, and that report calibrated confidence so responders know when to trust them.

Respond faster

Objective 3 → *addresses Challenge 3* → ***Case 6, 7***

Build place-based representations and accountable GeoAI agents that anticipate damage before an event, inspect it after, and deliver findings with evidence lineage and explicit refusal when evidence is insufficient.

Prepare and deliver better



STAGE 1 · OBJECTIVE 1

See better

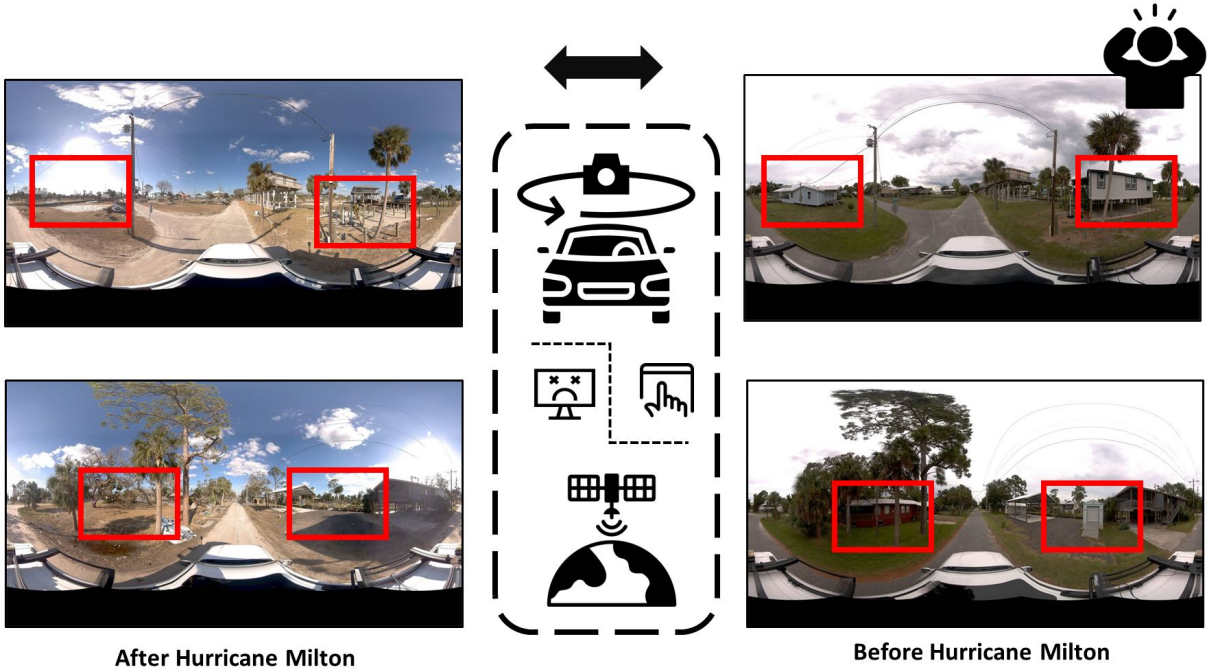
Challenge 1 · Place is only partially observed

Assess damage more completely by observing the same place across overhead, ground-level, and pre-disaster views, and by treating disagreement between views and models as evidence to arbitrate rather than noise to discard.

Case 1	Hyperlocal damage assessment from bi-temporal street-view imagery	*Published · CEUS 2025*
Case 2	DamageArbiter: multimodal arbitration for street-view damage assessment	*Under review · Natural Hazards*

Case 1:

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)



After Hurricane Milton

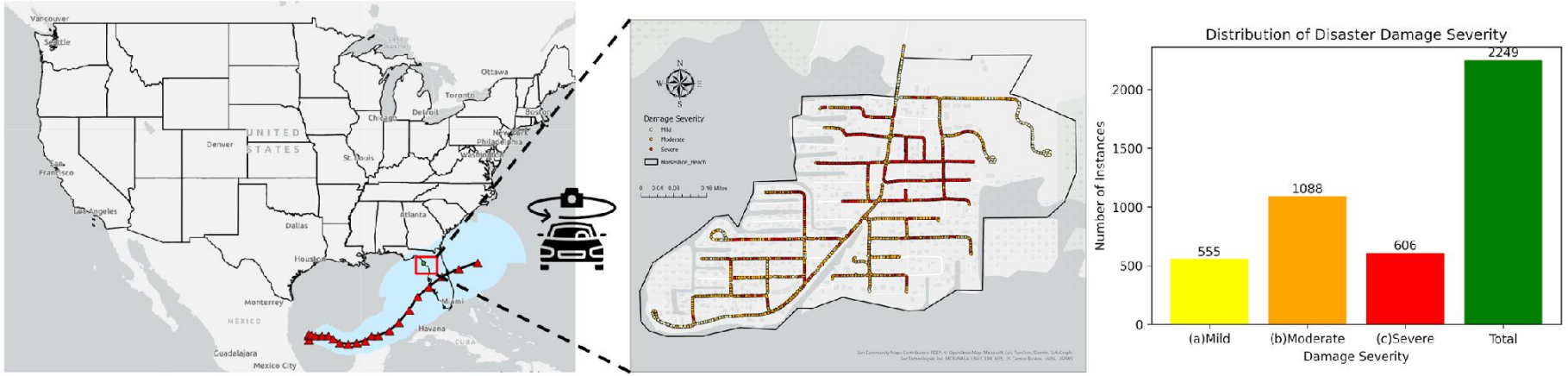
Before Hurricane Milton

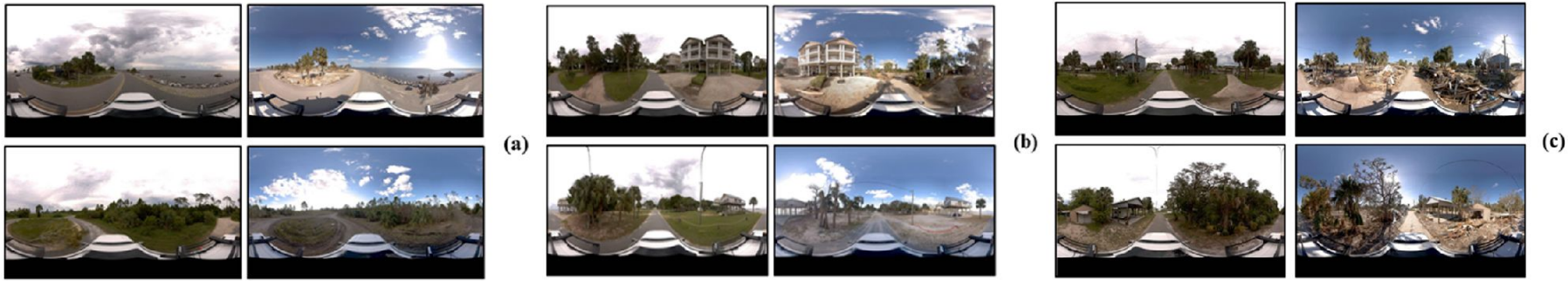
Research questions:

(1) Can pre-disaster street-view imagery help us classify disaster damage more accurately?

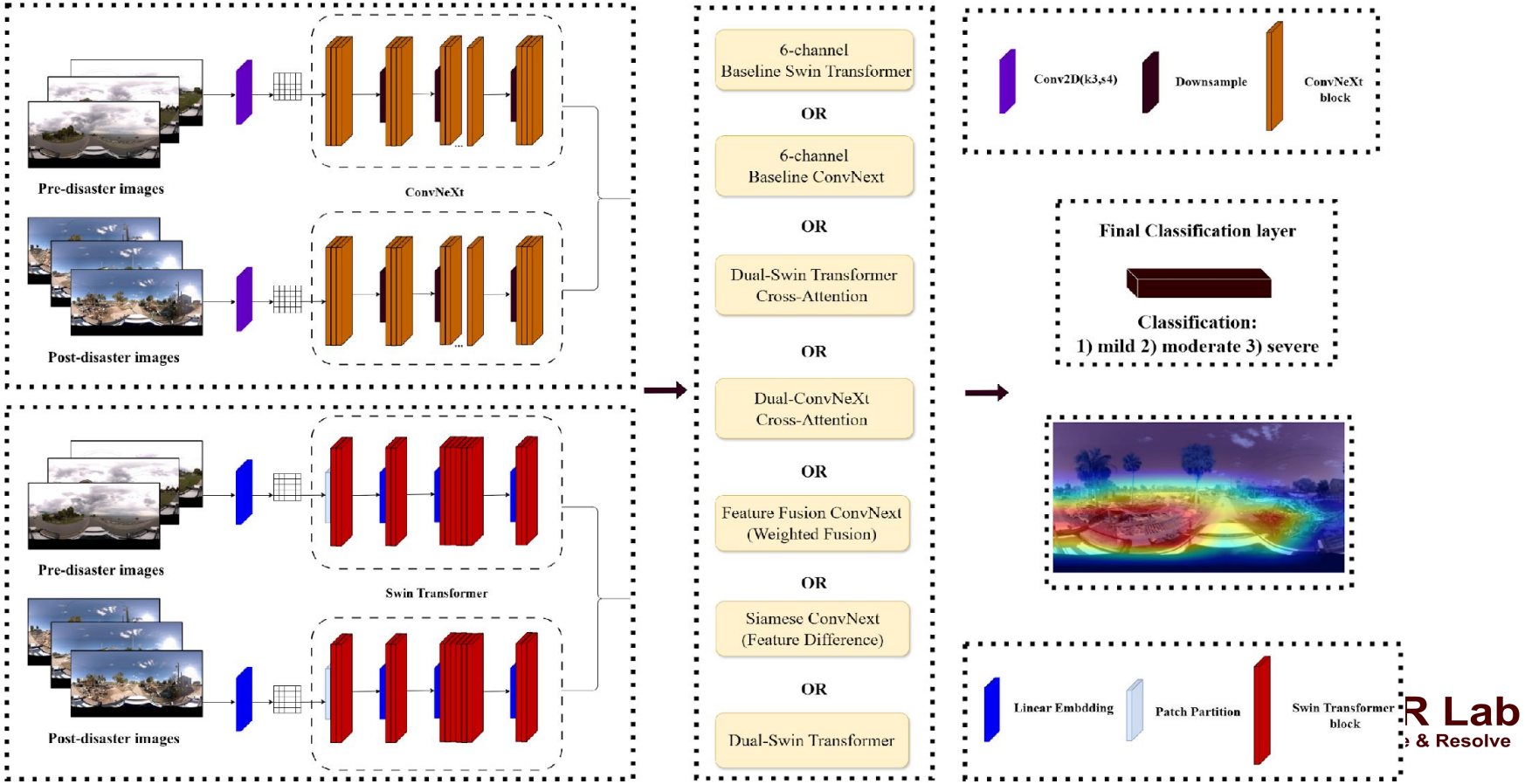
(2) How can we integrate pre- and post-disaster street-view imagery to enable rapid, hyperlocal damage assessment?

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (study area and dataset)

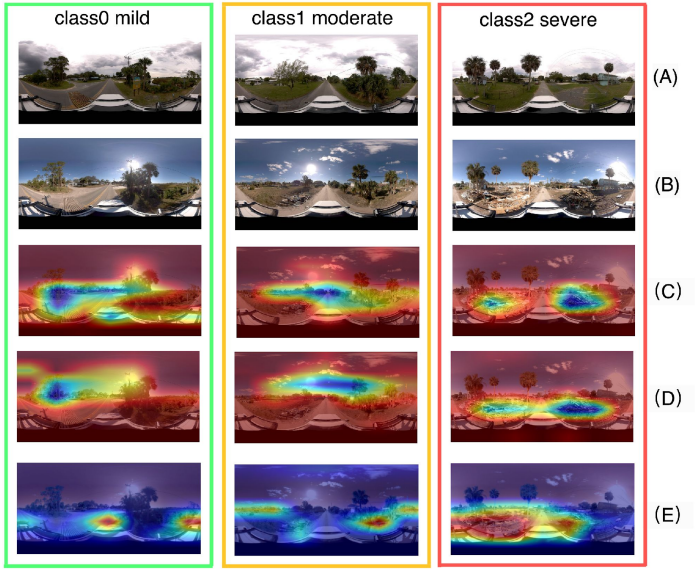




Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



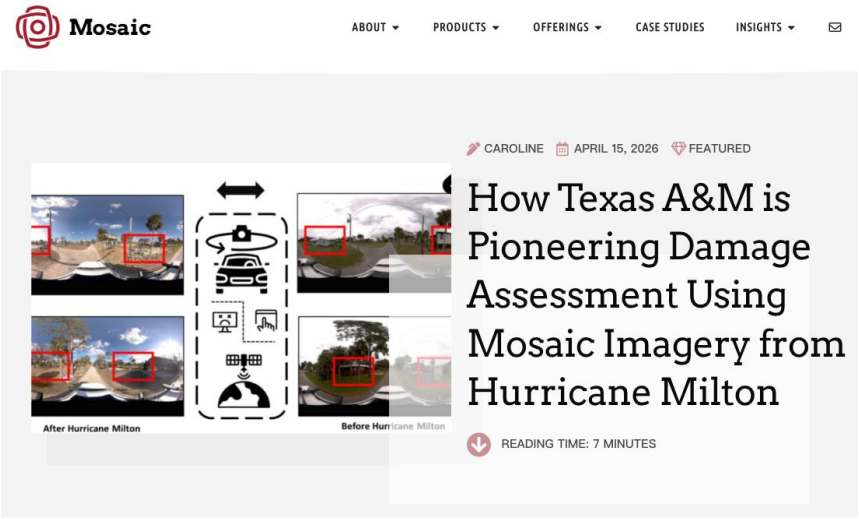
Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models



Model name	Accuracy	P	R	F1	Class-Specific Accuracy		
					0-mild	1-moderate	2-severe
Baseline-ConvNext	72.80%	0.7315	0.7280	0.7256	56.91%	79.12%	75.54%
Baseline-Swin Transformer	66.14%	0.6889	0.6614	0.6481	39.02%	84.34%	57.55%
Dual-ConvNeXt + Cross-Attention	76.91%	0.7704	0.7691	0.7672	***71.08%***	72.36%	***91.37%***
Dual-Swin Transformer + Cross-Attention	76.91%	***0.7970***	0.7691	0.7615	51.47%	***91.97%***	74.60%
Dual-Swin Transformer	74.17%	0.7420	0.7417	0.7408	62.69%	75.45%	81.33%
Feature-fusion ConvNext	***77.11%***	0.7751	***0.7710***	***0.7678***	59.39%	82.72%	84.44%
Siamese ConvNeXt	75.54%	0.7878	0.7554	0.7498	53.28%	90.79%	71.11%

Key result: Accuracy improved from 66.14% (Swin baseline) to 77.11% (dual-channel Feature-Fusion ConvNeXt); Grad-CAM confirms pre-disaster images help focus on true pre/post changes.

Case 1: Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models (Published, CEUS 2025)



https://www.mosaic51.com/featured/how-texas-am-is-pioneering-damage-assessment-using-mosaic-imagery-from-hurricane-milton/

Yang, Y., Zou, L., Zhou, B., Li, D., Lin, B., Abedin, J., & Yang, M. (2025). Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models. Computers, Environment and Urban Systems, 121, 102335.

Computers, Environment and Urban Systems 121 (2025) 102335

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journal homepage: www.elsevier.com/locate/ceus

Hyperlocal disaster damage assessment using bi-temporal street-view imagery and pre-trained vision models

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ARTICLE INFO

Keywords:
Disaster resilience
Street-view imagery
Dual-channel neural network
Pre-trained vision model
Damage estimation

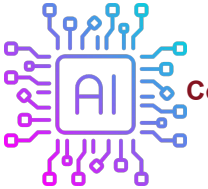
ABSTRACT

Street-view images offer unique advantages for disaster damage estimation as they capture impacts from a visual perspective and provide detailed, on-the-ground insights. Despite several investigations attempting to analyze street-view images for damage estimation, they mainly focus on using post-disaster images. The potential of time-series street-view images remains underexplored. Pre-disaster images provide valuable benchmarks for accurate damage estimations at building and street levels. These images could also aid annotators in objectively labeling post-disaster impacts, improving the reliability of labeled data sets for model training, and potentially enhancing the model performance in damage evaluation. The goal of this study is to estimate hyperlocal, on-the-ground disaster damages using bi-temporal street-view images and advanced pre-trained vision models. Street-view images before and after 2024 Hurricane Milton in Horseshoe Beach, Florida, were collected for experiments. The objectives are: (1) to assess the performance gains of incorporating pre-disaster street-view images as a no-damage category in fine-tuning pre-trained models, including Swin Transformer and ConvNeXt, for damage level classification; (2) to design and evaluate a dual-channel algorithm that reads pair-wise pre- and post-disaster street-view images for hyperlocal damage assessment. The results indicate that incorporating pre-disaster street-view images and employing a dual-channel processing framework can significantly enhance damage assessment accuracy. The accuracy improves from 66.14 % with the Swin Transformer baseline to 77.11 % with the dual-channel Feature-Fusion ConvNeXt model. Gradient-weighted Class Activation Mapping (Grad-CAM) shows that incorporating pre-disaster images improves the pre-trained vision model's capacity to focus on major changes between pre- and post-disaster images, thus enhancing model performance. This research evaluates the technical solutions and challenges of assessing disaster damages at hyperlocal spatial resolutions with bi-temporal street-view images, providing valuable insights to support effective decision-making in disaster management and resilience planning.

Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery (Under Review, Nature Hazard)

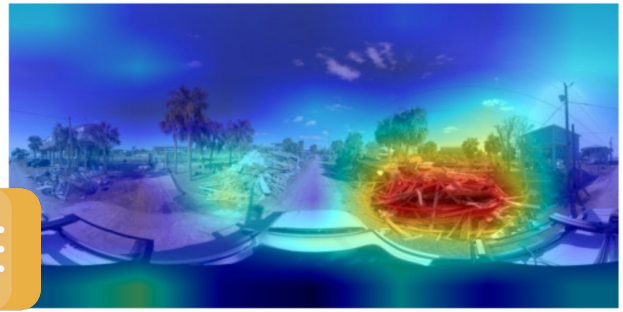


Post-Hurricane
Street-View Imagery



Computer Vision Models

Grad-CAM for Fine-Tuned Swin-Small





Motivation

- Accuracy alone is not enough
- Reliability matters for disaster response
- Overconfident wrong predictions
- Risk of misleading prioritization

Overconfidence=models may give very confident predictions even when they are wrong

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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

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class0 mild



No buildings in the area were damaged by the hurricane, and the trees appear to have sustained only mild damage. This is truly a relief. The vegetation is lush, the trees are tall, and the damage from the hurricane was minimal. I'd love to live here now, as it's only slightly damaged and looks very livable.

Sagging power lines are visible alongside a clear, dry road. Many trees show defoliation and some branch damage, but no large fallen trees or building debris are present. The overall scene suggests mild environmental impact, with infrastructure largely intact.

class1 moderate



The damage level is moderate, with loose building materials and a small amount of debris suggesting recent construction or minor disrepair. Despite the somewhat cluttered appearance of scattered items, the water features and elevated residences make this area appear to be an attractive residential area.

Scattered wooden debris and loose utility lines on a power pole indicate moderate damage along this coastal dirt road. While buildings appear structurally sound, the presence of scattered planks and beams, alongside the compromised power infrastructure, suggests the area experienced some impact, but not widespread destruction or flooding.

class2 severe



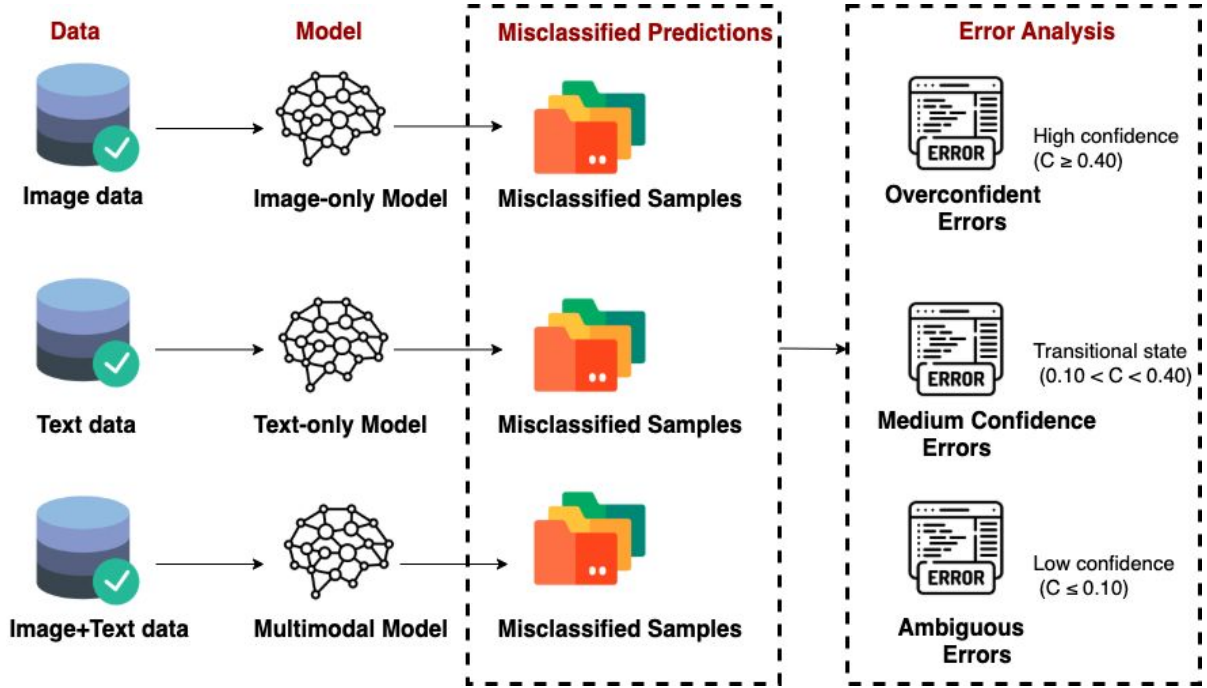
The atmosphere of the image is chaotic, with massive piles of debris covering large parts of the area, giving a strong sense of destruction. The damage appears extensive, with shattered building materials strewn all over, although some elevated houses remain intact in the background.

The street is heavily impacted by extensive building debris, forming large piles along both sides, almost completely blocking passage. This significant accumulation of scattered materials and collapsed fragments suggests severe damage to structures in the vicinity, though visible infrastructure like power lines appears mostly intact.

RQ1: How reliable are image-only, text-only, and image-text models for street-view disaster damage assessment?

RQ2: Can LLM-generated image descriptions support reliable damage assessment when human annotations are limited?

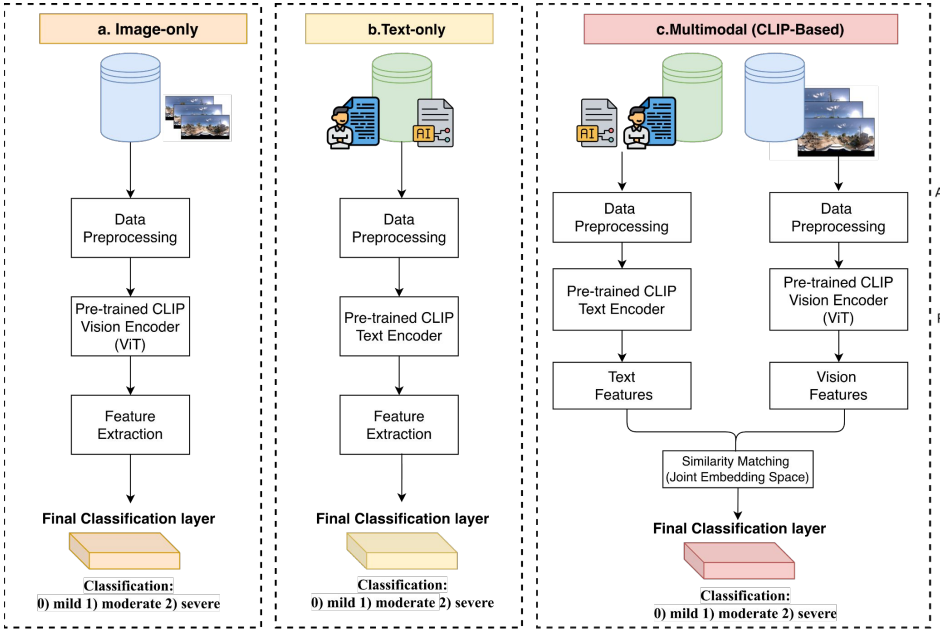
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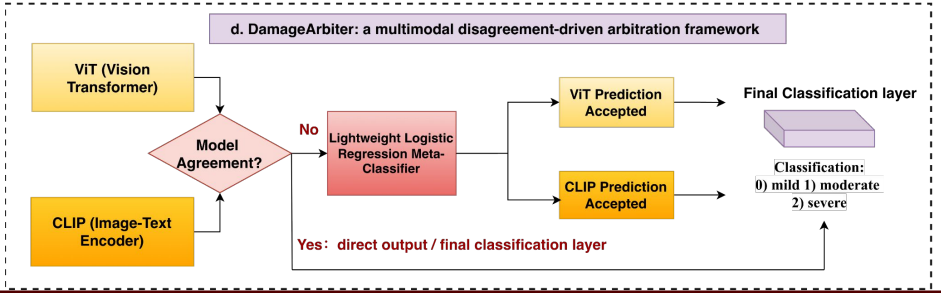


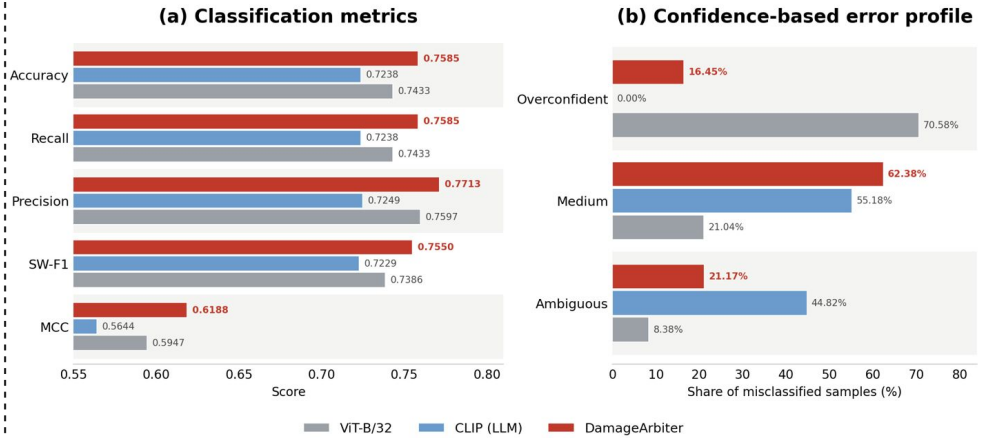
Overconfidence=models may give very confident predictions even when they are **wrong**

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Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

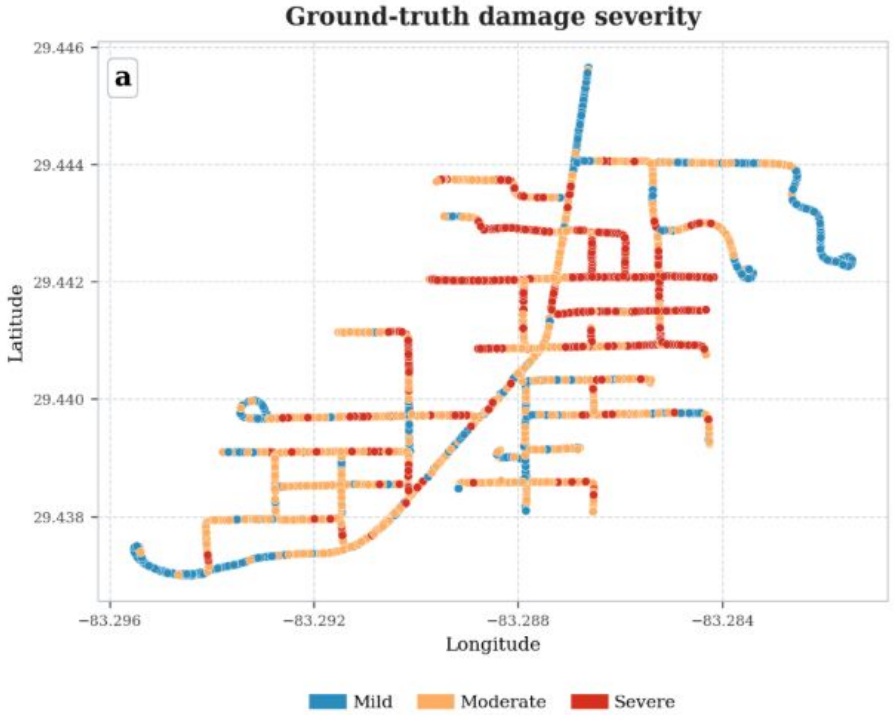


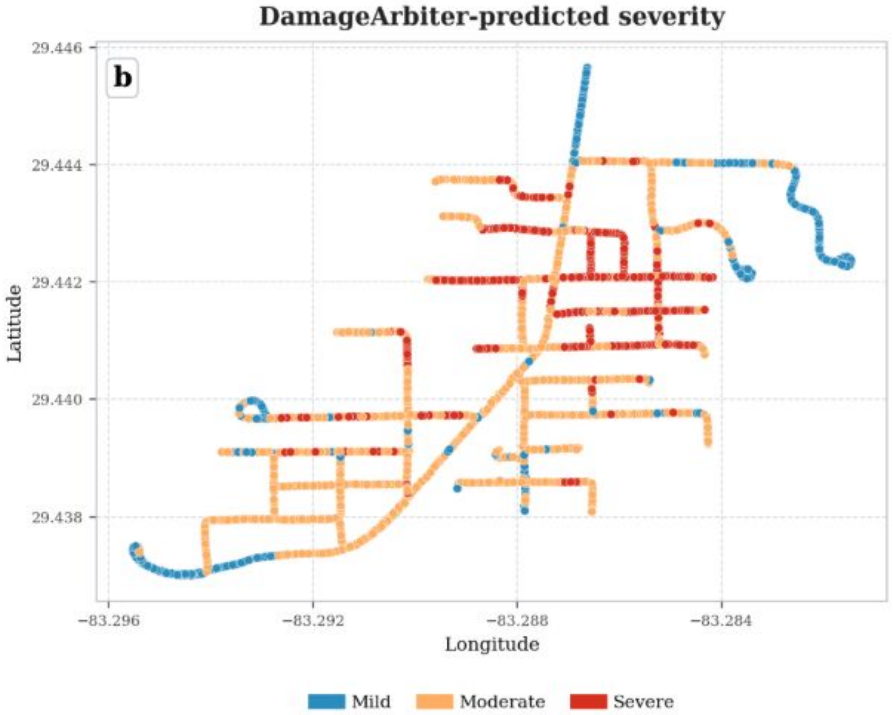




Case 2: DamageArbiter: A Multimodal Arbitration Framework for Disaster Damage Assessment from Street-View Imagery

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This work won 2nd Place in the 2026 AAG GIS Specialty Group (GISSG) Student Honors Paper Competition.

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STAGE 2 · OBJECTIVE 2

Respond faster

Challenge 2 · Rapid assessment without event-specific training

Develop damage assessment methods that transfer across events without new labels, retraining, or event-specific models, and that report calibrated confidence so responders know when to trust them.

Case 3	RAPID: multi-agent pipeline for interpretable cross-view assessment	*Accepted · ACM SIGSPATIAL 2026*
Case 4	Satellite-to-Street: post-disaster street views from satellite imagery	*Accepted · IEEE IGARSS 2026*
Case 5	Generative AI for disaster resilience	*Under review · CRC Press*

Case 3:

RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery (Accepted by ACM SIGSPATIAL)



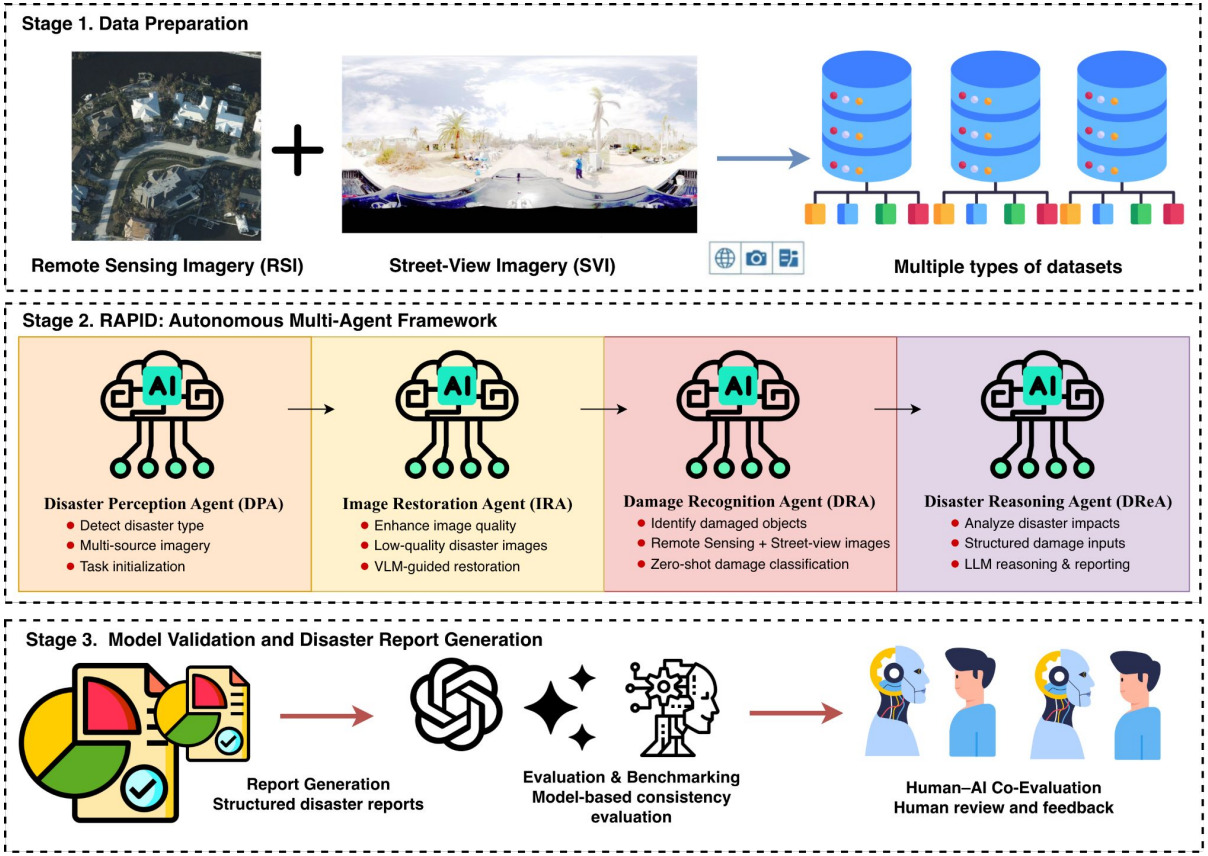
Research Question

Can RAPID support:

- Cross-view assessment?
- Different regions and hazards?
- Rapid zero-shot reasoning?
- Interpretable damage reports?



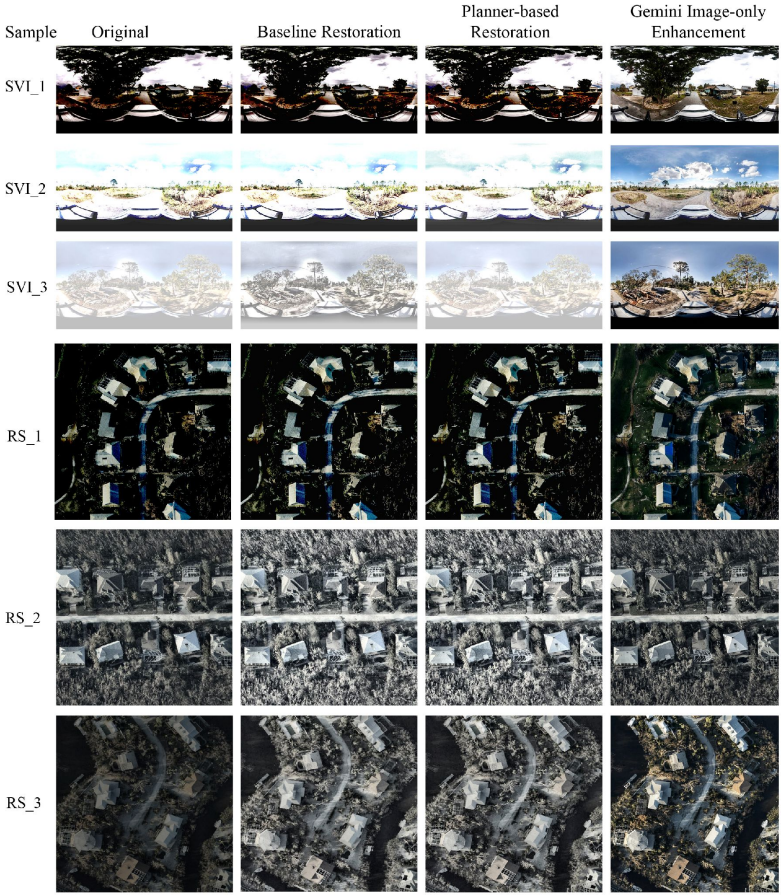
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery



Method:

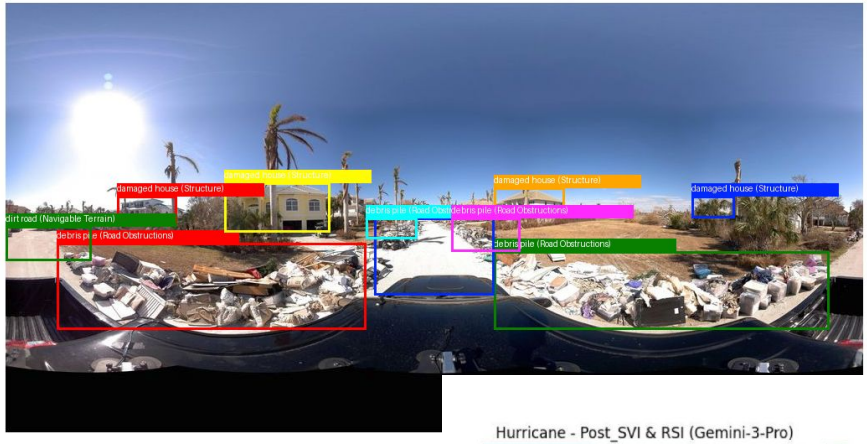
- **Cross-view imagery input**
- **Zero-shot multi-agent assessment**
- **Structured report generation**

Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery



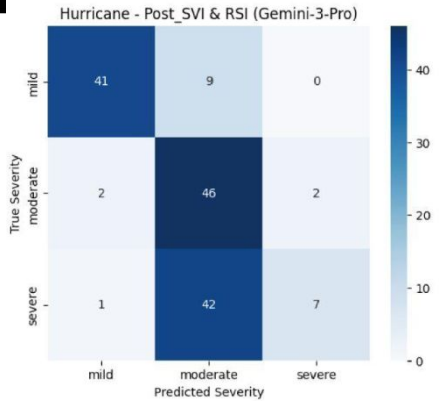
Disaster type	Image type	Restored images	$Q_{original}$	$Q_{baseline}$	$Q_{planner}$	Q_{gemini}
Hurricane	Satellite	141	0.62	**0.73**	0.71	0.69
Hurricane	SVI	22	0.75	0.78	0.76	**0.79**
Hurricane	SVI	4	0.76	0.78	0.78	**0.79**
Wildfire	SVI	14	0.61	**0.67**	**0.67**	0.57
Hurricane	SVI	2	0.51	0.55	**0.62**	0.60
Ice storm	SVI	4	0.73	0.74	**0.75**	0.67
Flood	SVI	2	0.68	0.73	**0.77**	0.71
Earthquake	SVI	2	0.71	0.79	**0.80**	0.77
Drought	SVI	2	0.66	0.75	**0.78**	0.61
Wildfire	SVI	3	0.70	0.77	**0.78**	**0.78**

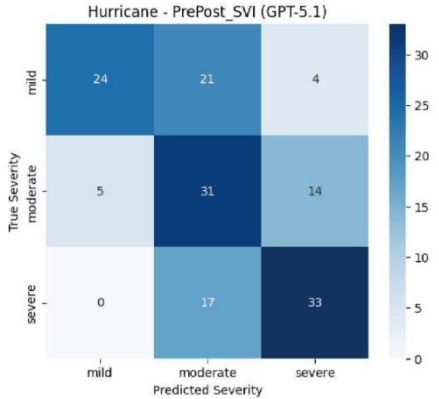
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

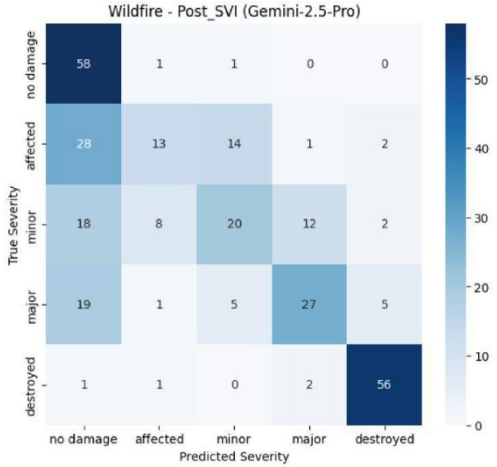


Metric	Model	Dataset A	Dataset B	Dataset C2
Accuracy ↑	GPT-5-mini	0.387	0.503	0.573
	ChatGPT-5.1	0.573	**0.591**	0.570
	Gemini-2.5-flash	0.360	0.470	0.559
	Gemini-2.5-Pro	0.380	0.447	**0.590**
	Gemini-3-Pro	**0.627**	0.493	0.442
NCSE ↓	GPT-5-mini	0.307	0.248	0.165
	ChatGPT-5.1	0.213	**0.218**	0.173
	Gemini-2.5-flash	0.363	0.299	0.179
	Gemini-2.5-Pro	0.373	0.327	**0.162**
	Gemini-3-Pro	**0.190**	0.291	0.210



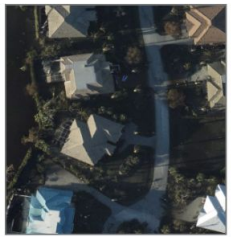
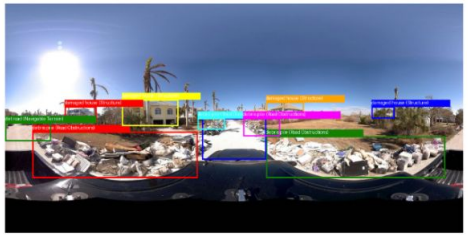






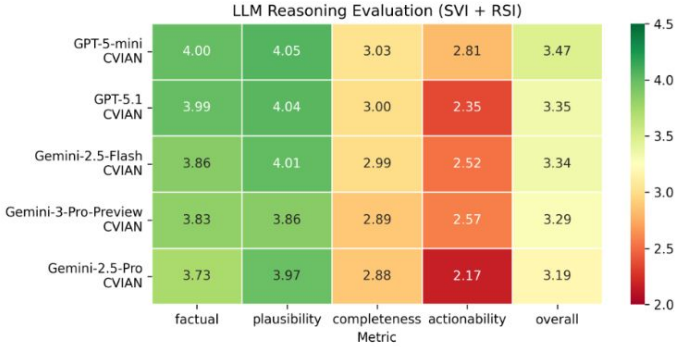
Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

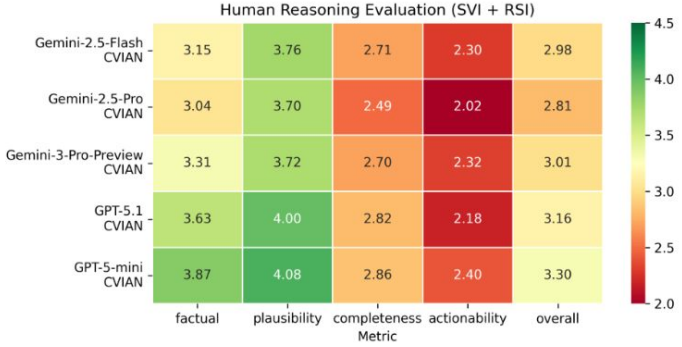
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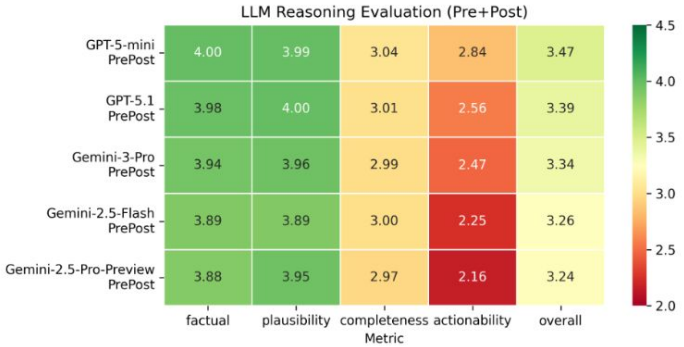
Input Images	Model Responses	Gemini-3-Pro	Gemini-2.5-Pro	ChatGPT-5.1
 	Type of disaster	hurricane	hurricane	hurricane
	Image restoration	No	No	No
	Disaster severity classification	Severe	Severe	severe
	Voted candidate	SVI and RSI	SVI and RSI	SVI and RSI
	Confidence	0.85	0.95	0.96
	Object recognition	Debris, Fallen Tree	Debris, Fallen Tree, Damaged Building	Debris, Fallen Tree, Damaged Building
 Using Gemini-3-Pro for object recognition	The Street View Image reveals massive piles of debris (furniture, appliances, drywall) lining the road, indicating that the adjacent homes were gutted by storm surge, which constitutes 'significant building debris' and major economic destruction. While the roads are not currently 'fully flooded' (they are covered in surge sand) and trees are mostly standing (though stripped), the sheer volume of property loss and the saturated ground visible in the Remote Sensing image align the overall impact with the 'Severe' classification. **Reasoning for Gemini-3-Pro-Disaster Assessment**			
	Recovery efforts must prioritize the rapid removal of the substantial debris (furniture, appliances, drywall) currently lining the streets to fully restore logistical access. Given that homes were 'gutted' by storm surge, reconstruction requires rigorous structural inspections and professional mold remediation before habitation is permitted. Infrastructure teams should focus on clearing surge sand from roadways and assessing subsurface utility stability due to the 'saturated ground.' Long-term planning should enforce elevated construction standards to mitigate future surge risks. **Reasoning for Gemini-3-Pro-Disaster Recovery / Reconstruction Suggestion**			

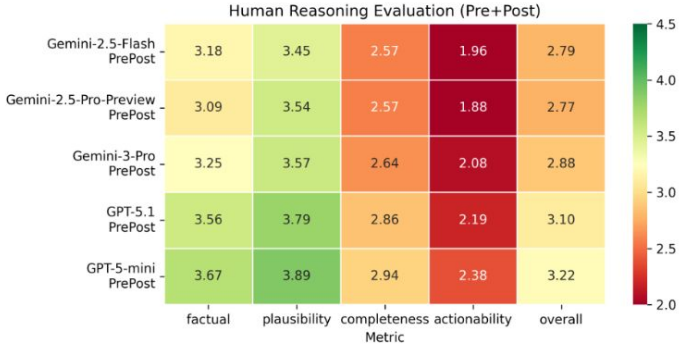
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Case 3: RAPID: A Reproducible Multi-Agent Pipeline for Interpretable Disaster Damage Assessment from Satellite and Street-View Imagery

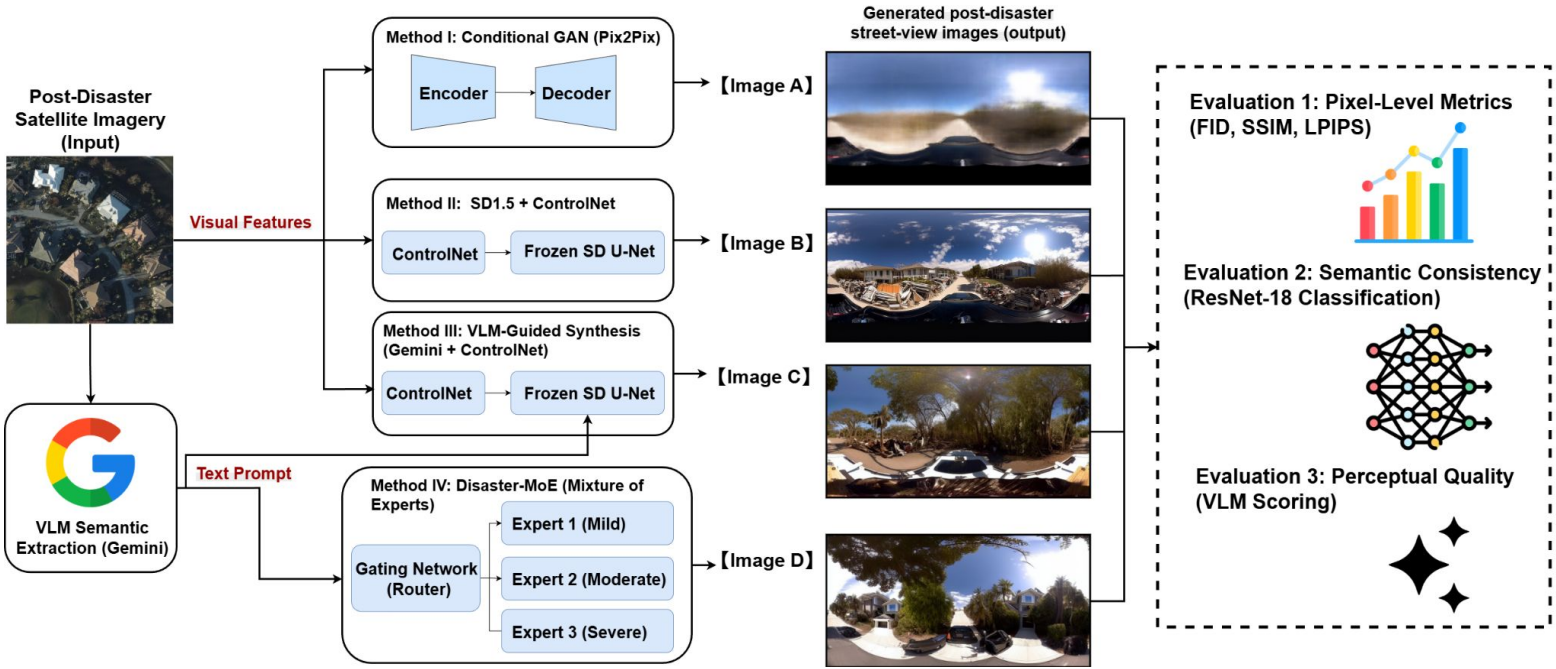






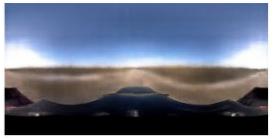
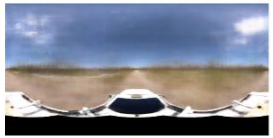
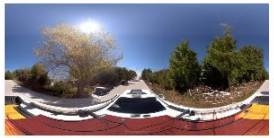


Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)



RQ1: Can generative vision models synthesize realistic post-disaster street views from satellite imagery?
RQ2: Do explicit disaster semantics improve synthesis fidelity?
RQ3: How can we reliably evaluate cross-view disaster synthesis beyond pixel metrics?

Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)

	Satellite	Pix2Pix	SD1.5+ControlNet	SD1.5+ControlNet+Gemini	MoE	Ground Truth
Mild						
Moderate						
Severe						

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Case 4: Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models (Accepted by IEEE International Geoscience and Remote Sensing Symposium)

Table 1 Quantitative Comparison of Perceptual Image Quality Metrics

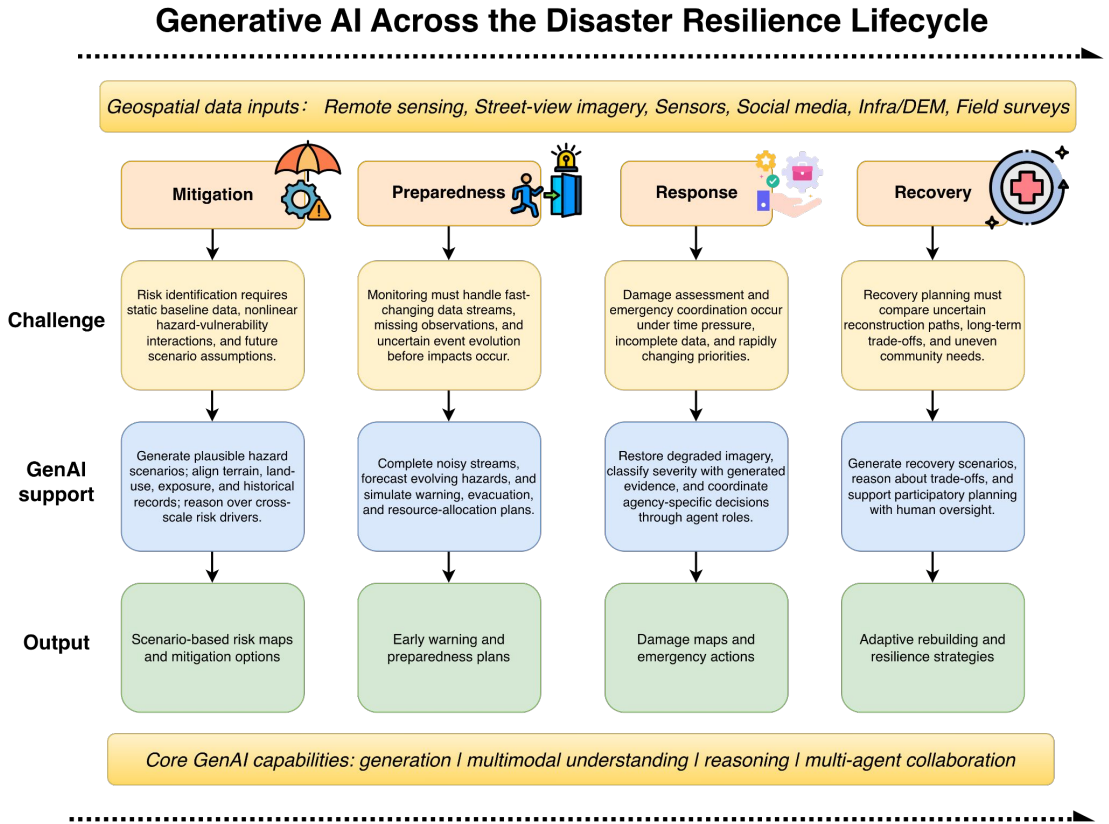
Method	SSIM(↑)	PSNR(↑)	LPIPS(↓)	FID(↓)
Pix2Pix	**0.58609614**	**15.30729573**	**0.549302574**	150.8341348
SD1.5_ControlNet	0.31436218	9.809302381	0.601944895	**74.32713682**
SD1.5_ControlNet_Gemini	0.290885676	9.726198367	0.603852578	82.19407033
MoE	0.221760725	8.453305609	0.6879829	134.522164
LoRA	0.1312	8.9619	0.6223	171.5578

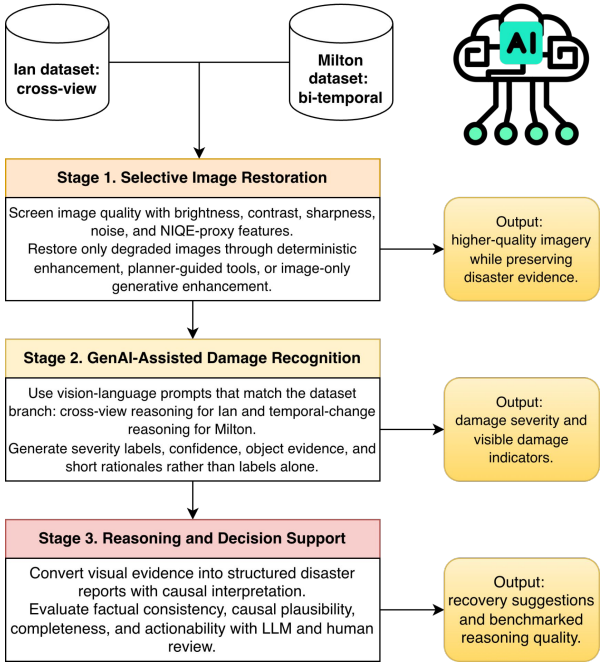
Table 2 Classification Accuracy Score for Disaster Severity Consistency

Method	Accuracy	Recall	Precision	F1_Score	Mild_Acc	Mod_Acc	Sev_Acc
GroundTruth	*0.73*	*0.73*	*0.76*	*0.74*	*0.77*	*0.76*	*0.66*
Pix2Pix	0.34	0.34	0.28	0.17	**1**	0.01	0
SD1.5_ControlNet	**0.72**	**0.72**	**0.74**	**0.71**	**0.91**	0.4	**0.86**
SD1.5_ControlNet_Gemini	0.43	0.43	0.45	0.43	0.4	0.39	0.5
MoE	0.43	0.43	0.45	0.44	0.41	**0.47**	0.42
LoRA	0.37	0.37	0.39	0.34	0.27	0.12	0.73

- Single-model approaches can't yet balance visual plausibility with strict structural alignment
- High perceptual realism ≠ high semantic accuracy — evaluation must be structure-aware
- Trustworthiness, not fidelity alone, should be the target for disaster generation

Case 5: Generative AI for Disaster Resilience
(Under Review, Geography in the Age of Generative AI (CRC Press))





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STAGE 3 · OBJECTIVE 3

Prepare and deliver better

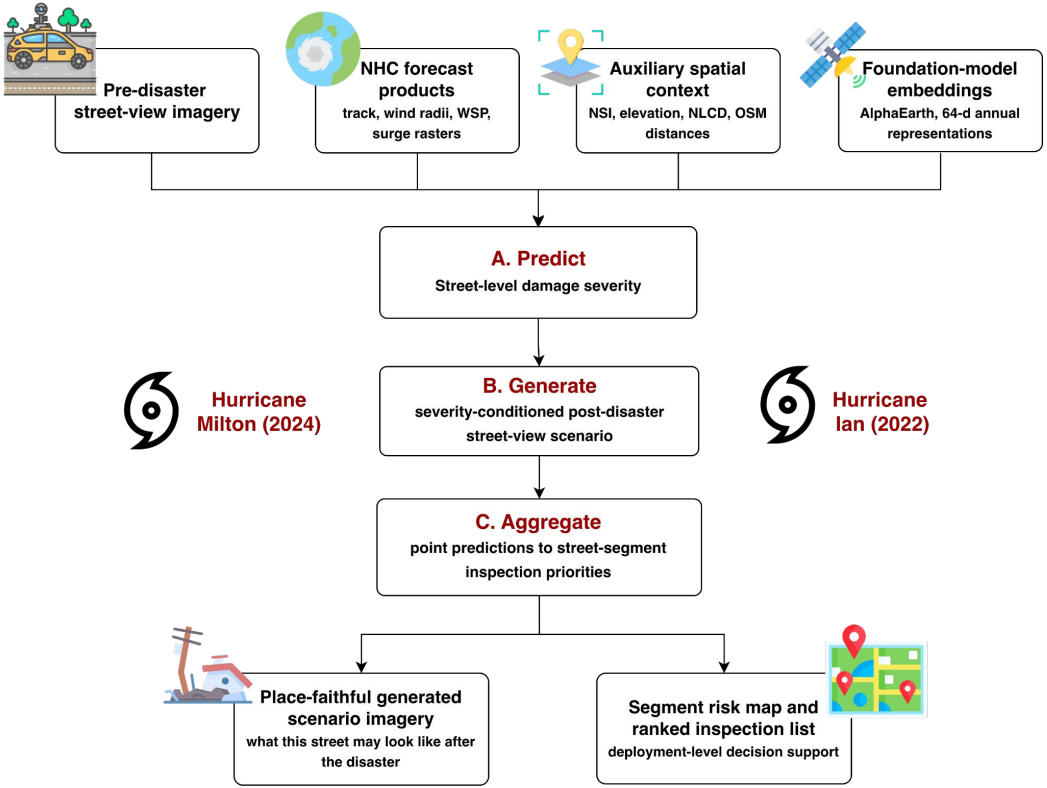
Challenge 3 · Place is difficult to anticipate, inspect, and communicate

Build place-based representations and accountable GeoAI agents that anticipate damage before an event, inspect it after, and deliver findings with evidence lineage and explicit refusal when evidence is insufficient.

Case 6	Seeing damage before it happens: anticipatory street-level prediction	*Ongoing*
Case 7	RAY: an accountable GeoAI system for place-based disaster intelligence	*Ongoing*

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Case 6: Seeing Disaster Damage Before It Happens: Predicting and Visualizing Street-Level Impacts from Multimodal Pre-Event Information (Ongoing)



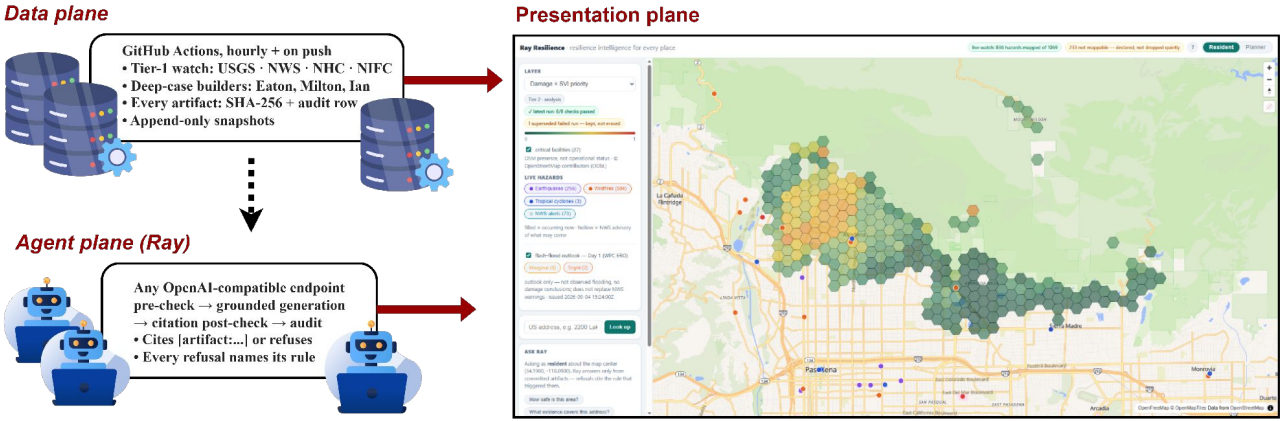
Research Question:

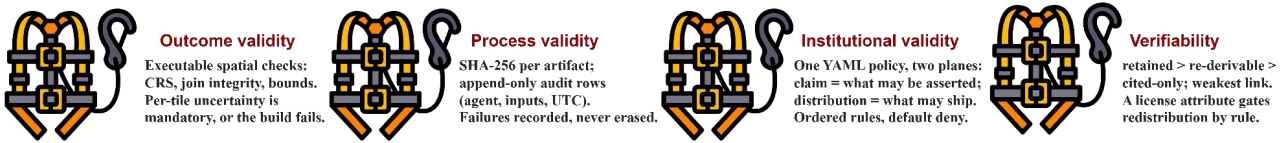
- **Can we anticipate post-disaster street-level damage using pre-disaster multi-modal data, Earth observation embeddings, and hazard forecasts?**

Objective:

- **To develop a preparedness-oriented GeoAI framework that transforms pre-disaster data and forecasts into anticipatory damage predictions, spatial risk prioritization, and scenario visualizations.**

Case 7: RAY: Resilience Assistant for You
An Accountable GeoAI System for Place-Based Disaster Intelligence (Ongoing)





Research Question:

How can an **automated, multi-agent GeoAI system** deploy large models within WebGIS and mobile apps to deliver accountable, evidence-backed, and real-time resilience awareness for both the public and decision-makers?

Objective:

To implement an **accountable multi-agent GeoAI architecture**—such as the RAY system—that transforms complex spatial data into real-time, interactive insights featuring explicit evidence lineage, planning trade-off evaluation, and differentiated multi-tier dissemination.

Publications and Awards

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As of October 2026 · Google Scholar: 429 citations, h-index 8, i10-index 7.

PUBLICATIONS

8 first-authored publications during Ph.D.

- 2 journal articles (CEUS 2025; Natural Hazards under review)
- 5 conference proceedings (ACM SIGSPATIAL 2026; ACM SIGSPATIAL 2026 Workshop GeoSearch 26; IEEE IGARSS 2026; CaGIS 2026; ICC 2025)
- 1 chapter under review (CRC Press)

15 co-authored publications

- 9 journal articles (e.g., Journal of Hazardous Materials, Int. J. Applied Earth Observation and Geoinformation, Urban Informatics, Journal of Transport Geography)
- 3 conference papers (ACM SIGSPATIAL 2026 short paper; AGU 2025) and 3 preprints



HONORS AND AWARDS

A total of 15 honors received during Ph.D.

2026

- 2nd Place, AAG GISSG Student Honors Paper Competition
- AAG International Geographic Information Fund (IGIF) Scholarship
- CaGIS Doctoral Scholarship Award
- IGARSS 2026 Travel Grant
- NSF I-GUIDE Summer School Travel Award
- ACM SIGSPATIAL 2026 Travel Award
- Geography Graduate Student Enhancement Scholarship, Department of Geography, etc.

2025

- Best Student Paper Award, International Cartographic Conference (ICC 2025)
- Environment and Sustainability Graduate Fellow Award, TAMU
- TAMIDS Student Ambassador Scholarship
- CaGIS International Travel Grant, etc.

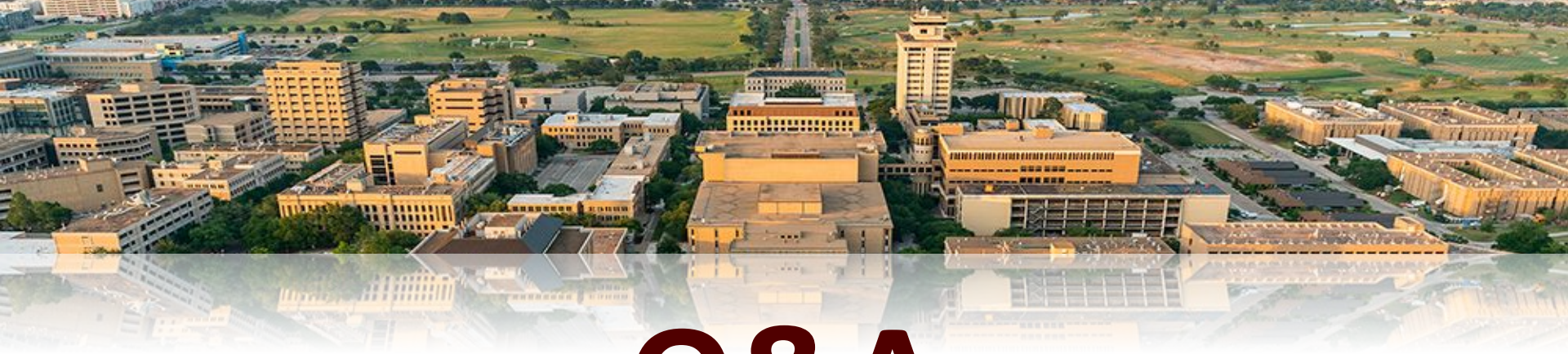
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Research Schedule

Now

Tasks	Fall 2024	Spring 2025	Summer 2025	Fall 2025	Spring 2026	Summer 2026	Fall 2026	Spring 2027	Summer 2027
Case Study 1	Done								
Case Study 2		Done							
Case Study 3				Done					
Case Study 4					Done				
Case Study 5					Done				
Case Study 6							In Progress (20%)		
Case Study 7							In Progress (40%)		
Prelim Exam					Done				
Proposal Defense							In Progress		
Dissertation Writing							In Progress		
Final Defense & Graduation									Planned

Done In Progress Planned



Q&A

Welcome any comments/questions

yyang295@tamu.edu

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